



Chapter 4: Motion in 2 dimensions

Motion in a plane – 2D

Average Velocity

Average Acceleration

Projectile motion

Circular motion

Position

Velocity

Acceleration

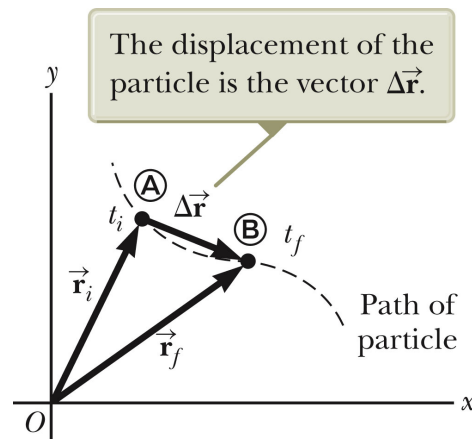
General Motion Ideas

► In two- or three-dimensional kinematics, everything is the same as in one-dimensional motion except that we must now use full vector notation.

- Positive and negative signs are no longer sufficient to determine the direction.

Position and Displacement Vectors

- ▶ The **position** of an object is described by its position vector, $\vec{r}$.
 $\vec{r} = x\hat{i} + y\hat{j}$
- ▶ The **displacement** of the object is defined as the *change in its position*.
 $\Delta\vec{r} \equiv \vec{r}_f - \vec{r}_i$
- ▶ Independent of path



Section 4.1

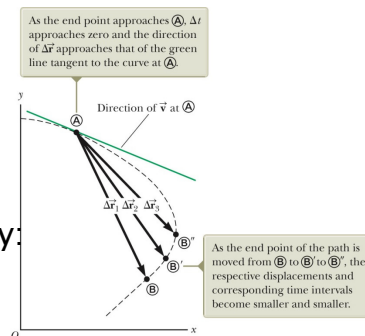
Average and Instantaneous Velocity

- ▶ Following the same approach as in Ch 2 we define **average velocity**:

$$\begin{aligned}\vec{v}_{avg} &= \frac{\Delta\vec{r}}{\Delta t} = \frac{\Delta r_x \hat{i} + \Delta r_y \hat{j} + \Delta r_z \hat{k}}{\Delta t} \\ &= \frac{\Delta r_x}{\Delta t} \hat{i} + \frac{\Delta r_y}{\Delta t} \hat{j} + \frac{\Delta r_z}{\Delta t} \hat{k}\end{aligned}$$

- ▶ **Instantaneous velocity** given by:

$$\vec{v} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$



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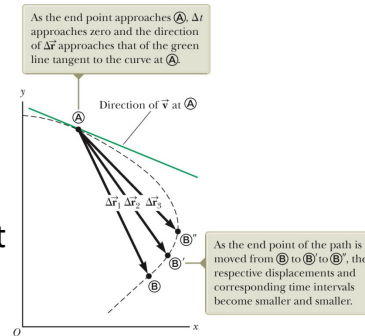
Average and Instantaneous Velocity

- ▶ Instantaneous velocity given by:

$$\begin{aligned}\vec{v} &= \frac{d\vec{r}}{dt} = \frac{d(x\hat{i} + y\hat{j} + z\hat{k})}{dt} \\ &= \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \\ &= v_x\hat{i} + v_y\hat{j} + v_z\hat{k}\end{aligned}$$

- ▶ Direction of $\vec{v}$ is always tangent to the particles path.
- ▶ Scalar components:

$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt} \quad v_z = \frac{dz}{dt}$$



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Example

The figure shows a circular path taken by a particle. If the instantaneous velocity of the particle is $\vec{v} = (2 \text{ m/s})\hat{i} - (2 \text{ m/s})\hat{j}$ through which quadrant is the particle moving at that instant if it is travelling (a) clockwise and (b) counter-clockwise around the circle? For both cases, draw on the figure.

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Average and Instantaneous Acceleration

- When a particle's velocity changes in a time interval Δt the **average acceleration** is:

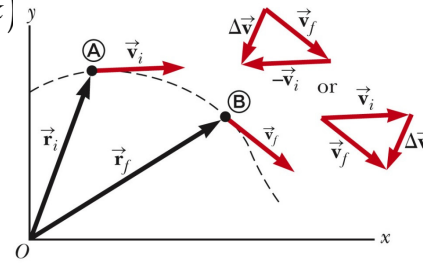
$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}$$

- As previously discussed the **instantaneous acceleration** is then:

$$\begin{aligned}\vec{a} &\equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d}{dt} (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) \\ &= \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k} \\ &= a_x \hat{i} + a_y \hat{j} + a_z \hat{k}\end{aligned}$$

- Scalar components:

$$a_x = \frac{dv_x}{dt} \quad a_y = \frac{dv_y}{dt} \quad a_z = \frac{dv_z}{dt}$$



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Examples

- Here are four descriptions of the position (in meters) of a puck as it moves in an xy plane:

1. $x = -3t^2 + 4t - 2$ and $y = 6t^2 - 4t$

2. $x = -3t^3 - 4t$ and $y = -5t^2 + 6$

3. $\vec{r} = 2t^2 \hat{i} - (4t + 3) \hat{j}$

4. $\vec{r} = (4t^3 - 2t) \hat{i} + 3 \hat{j}$

Are the x and y acceleration components constant? Is the acceleration constant?

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Example

A particle with velocity $\vec{v} = -2.0\hat{i} + 4.0\hat{j}$ (in meters per second) at $t = 0$ undergoes a constant acceleration $\vec{a}$ of magnitude $a = 3.0 \text{ m/s}^2$ at an angle $\theta = 130^\circ$ from the positive direction of the x axis. What is the particle's velocity $\vec{v}$ at $t = 5 \text{ s}$?

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Producing An Acceleration

- ▶ Various changes in a particle's motion may produce an acceleration.
 - The magnitude of the velocity vector may change.
 - The direction of the velocity vector may change.
 - Even if the magnitude remains constant
 - Both may change simultaneously

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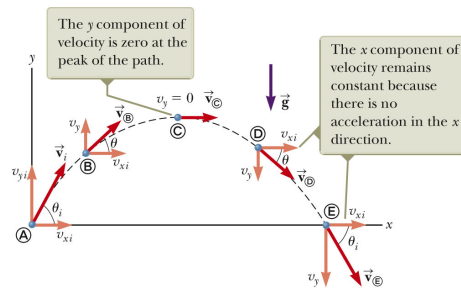
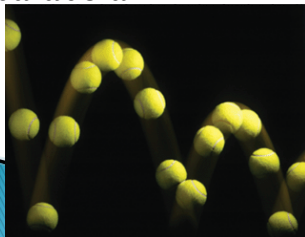
2-D motion with constant acceleration

- ▶ Position $\vec{r} = x\hat{i} + y\hat{j}$
- ▶ Velocity
$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(x\hat{i} + y\hat{j})}{dt}$$
$$= \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} = v_x\hat{i} + v_y\hat{j}$$
- ▶ Substituting $v_{xf} = v_{xi} + a_x t$ and $v_{yf} = v_{yi} + a_y t$
- ▶ $\vec{v}_f = (v_{xi} + a_x t)\hat{i} + (v_{yi} + a_y t)\hat{j}$ similarly $\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2}\vec{a}t^2$
$$= (v_{xi}\hat{i} + v_{yi}\hat{j}) + (a_x\hat{i} + a_y\hat{j})t$$
$$\vec{v}_f = \vec{v}_i + \vec{a}t$$

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Projectile Motion

- ▶ Special case: 2-D motion
- ▶ **Projectile motion**: particle moves in a vertical plane with an initial velocity $\vec{v}_0$ but acceleration is always the free fall of gravity downwards $\vec{a} = -\vec{g} = -9.8 \text{ m/s}^2$
- ▶ Assume no air resistance
- ▶ Path is shown
- ▶ Trajectory is always parabola



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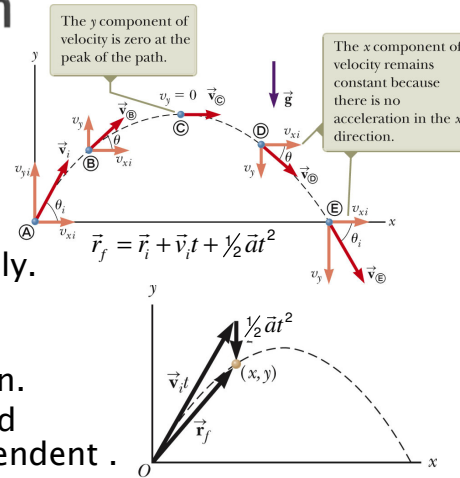
Projectile motion

$$\vec{v}_i = v_{ix} \hat{i} + v_{iy} \hat{j}$$

$$v_{ix} = v_i \cos \theta$$

$$v_{iy} = v_i \sin \theta$$

- ▶ Position vector + velocity vector change continuously.
- ▶ Acceleration is ALWAYS constant, downwards
- ▶ NO horizontal acceleration.
- ▶ The horizontal motion and vertical motion are independent .
- ▶ Consider the motion as the super-position of the motions in the x - and y -directions.
- ▶ Solve the two dimensions separately.



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Projectile Motion Analysed

- ▶ **Launch point:** x_i and y_i
 - Can be at the origin (but not necessarily)
- ▶ **Horizontal motion:**
 - No acceleration
 - Displacement: $x_f - x_i = v_{ix} t = (v_i \cos \theta) t$
- ▶ **Vertical motion:**
 - Acceleration is constant: $\vec{a} = -\vec{g} = -9.8 \text{ m/s}^2$
 - Can use equations from ch 2
 - Displacement $y - y_i = v_{iy} t + \frac{1}{2} a t^2$

$$= (v_i \sin \theta) t + \frac{1}{2} (-g) t^2 = (v_i \sin \theta) t - \frac{1}{2} g t^2$$
 - Other equations $v_{yf} = v_i \sin \theta - g t$

$$v_{yf}^2 = (v_i \sin \theta)^2 - 2 g \Delta y$$

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Projectile Motion Analysed

- ▶ Max height: h
- ▶ Range: R
- ▶ Time to reach h

$$v_{yf} = v_{yi} - gt$$

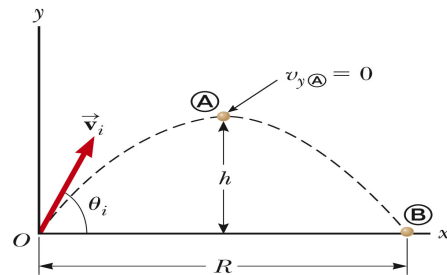
$$0 = v_i \sin \theta_i - gt$$

$$t = \frac{v_i \sin \theta_i}{g}$$

- ▶ Substituting this t into $y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$

$$h = 0 + v_i \sin \theta_i \left(\frac{v_i \sin \theta_i}{g} \right) - \frac{1}{2}g \left(\frac{v_i \sin \theta_i}{g} \right)^2$$

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}$$



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Projectile Motion Analysed

- ▶ Time of flight $t_R = 2t_h = \frac{2v_i \sin \theta}{g}$

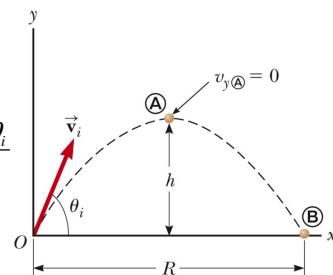
- ▶ Range:
- ▶ Horizontal velocity is constant

$$x_f = x_i + v_{xi}t$$

$$R = 0 + (v_i \cos \theta_i) \left(\frac{2v_i \sin \theta_i}{g} \right) = \left(\frac{2v_i^2 \sin \theta_i \cos \theta_i}{g} \right)$$

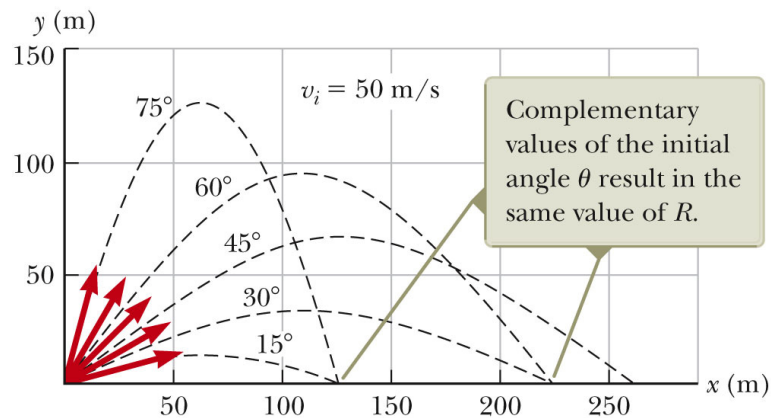
$$R = \frac{v_i^2 \sin 2\theta_i}{g}$$

- ▶ Max R when $\theta = 45^\circ$



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Projectile Motion Analysed



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Problem solving strategy

- ▶ Conceptualize
- ▶ Categorize – Choose axes (x – y), free fall (*vertical*)? Constant velocity (*horizontal*)
- ▶ Analyse – separate x and y variables and components of variables
- ▶ Finalize – check to see if your answers are consistent with what should happen – is it realistic?

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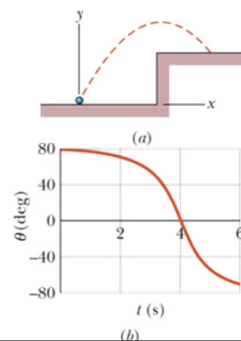
Problem

- ▶ A rifle with a muzzle velocity $v_0 = 450 \text{ m/s}$ shoots a bullet at a target 50 m away. How high above the target must the barrel be pointed? Assume the muzzle and the target are at the same height.

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Problem

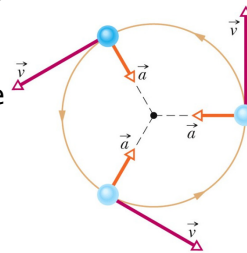
- ▶ At time $t = 0$ a golf ball is shot from ground level into the air, as shown in the fig. The angle θ between the ball's direction of travel and the positive x axis is given as a function of time t . The ball lands at $t = 6.00 \text{ s}$.
 - What is the magnitude v_0 of the ball's launch velocity?
 - At what height $(y - y_0)$ above the launch level does the ball land?
 - What is the ball's direction of travel just before it lands?



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Uniform Circular Motion

- ▶ A particle is in **uniform circular motion** if:
 - It travels in a circular arc
 - At a constant speed.
- ▶ Although speed is constant, there is acceleration
 - Velocity is changing direction
 - Velocity is directed tangent to the circle
 - Acceleration is directed to centre of the circle – centripetal acceleration.
 - If there was a component parallel to the speed, it would increase – not uniform circular motion



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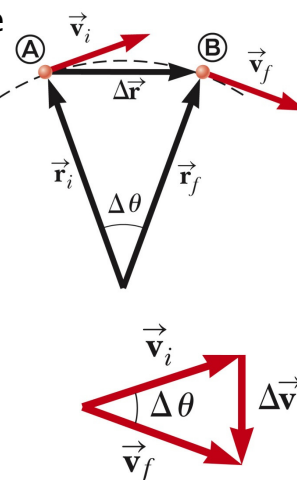
Uniform Circular Motion

- ▶ Consider the diagram with a particle moving from A to B on circular arc:

- Position vector, displacement in time Δt .
- Velocity vectors at A and B, and $\Delta \vec{v}$
- Positions triangle *similar* to velocities triangle.
- $\Delta \theta$ between the positions = $\Delta \theta$ between the velocity vectors.

$$\frac{|\Delta \vec{v}|}{v} = \frac{|\Delta \vec{r}|}{r}, \text{ where } v = v_i = v_f, r = r_i = r_f$$

$$|\vec{a}_{avg}| = \frac{|\Delta \vec{v}|}{|\Delta t|} = \frac{v |\Delta \vec{r}|}{r \Delta t}$$



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Uniform Circular Motion

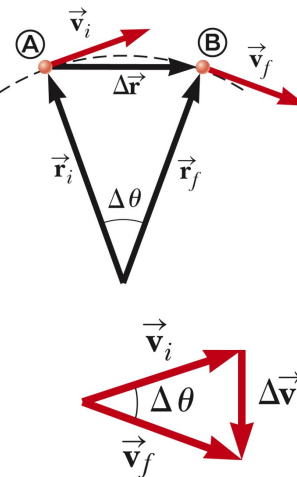
$$|\vec{a}_{avg}| = \frac{|\Delta \vec{v}|}{|\Delta t|} = \frac{v |\Delta \vec{r}|}{r \Delta t}$$

- As $\Delta t \rightarrow 0$, $|\Delta \vec{r}| \rightarrow$ distance on circular path.

$$\frac{|\Delta \vec{r}|}{\Delta t} \rightarrow v$$

$$\vec{a}_c = \frac{v^2}{r}$$

- centripetal acceleration** (centre-seeking)



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Uniform Circular Motion

- Period (T)** = time to complete one revolution

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\therefore T = \frac{\text{distance}}{\text{speed}} = \frac{2\pi r}{v}$$

- Angular speed ω = revolutions per second
- SI units are Radians/s (ClickUP video to convert from $^\circ$ to radians)

$$\omega = \frac{2\pi}{T}$$

- Combining these two equations: $\omega = 2\pi \left(\frac{v}{2\pi r} \right) = \frac{v}{r}$

$$v = r\omega$$

- Substituting into acceleration: $a_c = \frac{v^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2$

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Example

- ▶ An object moves at constant speed along a circular path in a horizontal xy plane, with the centre at the origin. When the object is at $x = -2\text{m}$, its velocity is $-(4\text{m/s})\hat{j}$. Give the object's (a) velocity and (b) acceleration at $y = 2\text{ m}$.

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Problem

- ▶ A satellite is at an altitude $h = 200\text{ km}$ above the earth ($g = 9.2\text{ m.s}^2$). Determine the satellite's velocity and its period.
The earth's radius is $6.37 \times 10^6\text{ m}$.

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Relative Motion

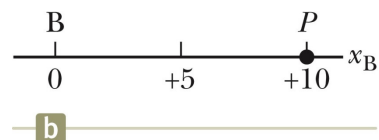
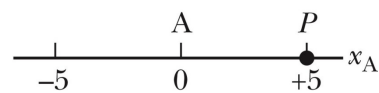
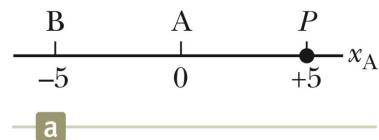
- ▶ Two observers moving relative to each other generally do not agree on the outcome of an experiment.
- ▶ However, the observations seen by each are related to one another.
- ▶ A frame of reference can be described by a Cartesian coordinate system for which an observer is at rest with respect to the origin.

Section 4.6

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Different Measurements example

- ▶ Observer A measures point P at +5 m from the origin
- ▶ Observer B measures point P at +10 m from the origin
- ▶ The difference is due to the different frames of reference being used.
- ▶ B moving to the right ...

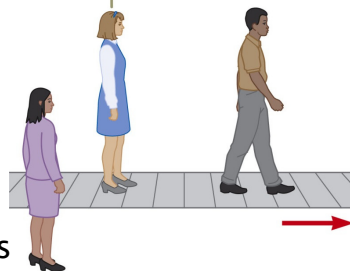


Section 4.6

Different Measurements example 2

- ▶ The man is walking on the moving beltway.
- ▶ The woman on the beltway sees the man walking at his normal walking speed.
- ▶ The stationary woman sees the man walking at a much higher speed.
 - The combination of the speed of the beltway and the walking.
- ▶ The difference is due to the relative velocity of their frames of reference.

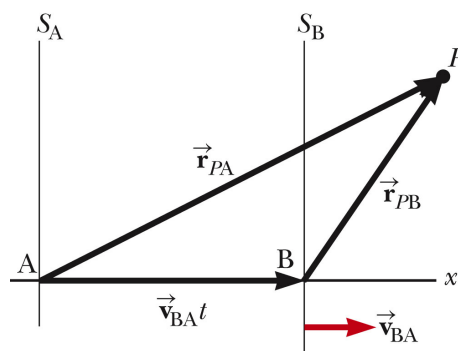
The woman standing on the beltway sees the man moving with a slower speed than does the woman observing the man from the stationary floor.



Section 4.6

Relative Velocity, generalized

- ▶ Reference frame S_A is stationary
- ▶ Reference frame S_B is moving to the right relative to S_A at $\vec{v}_{BA}$
 - This also means that S_A moves at $\vec{v}_{AB} = -\vec{v}_{BA}$ relative to S_B
- ▶ Define time $t = 0$ as that time when the origins coincide
- ▶ At time t they are separated by $v_{BA}t$



Section 4.6

Notation

- ▶ The first subscript represents what is being observed.
- ▶ The second subscript represents who is doing the observing.
- ▶ Example $\vec{v}_{BA}$
 - The velocity of B (and attached to frame S_B) as measured by observer A

Section 4.6

Relative Velocity, equations

- ▶ The positions as seen from the two reference frames are related through the velocity
 - $\vec{r}_{PA} = \vec{r}_{PB} + \vec{v}_{BA}t$
- ▶ The derivative of the position equation will give the velocity equation $\vec{u}_{PA} = \vec{u}_{PB} + \vec{v}_{BA}$
 - - $\vec{u}_{PA}$ is the velocity of the particle P measured by observer A
 - $\vec{u}_{PB}$ is the velocity of the particle P measured by observer B
- ▶ These are called the **Galilean transformation equations**.

Section 4.6

Acceleration in Different Frames of Reference

- ▶ The derivative of the velocity equation will give the acceleration equation.
- ▶ The acceleration of the particle measured by an observer in one frame of reference is the same as that measured by any other observer moving at a ***constant velocity*** relative to the first frame.

Section 4.6

Acceleration, cont.

- ▶ Calculating the acceleration gives

$$\frac{d\vec{u}_{PA}}{dt} = \frac{d\vec{u}_{PB}}{dt} + \frac{d\vec{v}_{BA}}{dt}$$

- ▶ Since

$$\vec{v}_{BA} \text{ is constant, } d\vec{v}_{BA}/dt = 0$$

- ▶ Therefore, $\vec{a}_{PA} = \vec{a}_{PB}$

Section 4.6

Problem

- Suppose that Bob's velocity relative to Alex is a constant $v_{BA} = 52 \text{ km/h}$ and car P is moving in the negative direction of the x axis. (a) If Alex measures a constant $v_{PA} = -78 \text{ km/h}$ for car P , what velocity v_{PB} will Bob measure? (b) If car P brakes to stop relative to Alex in time $t = 10 \text{ s}$ at constant acceleration, what is its acceleration a_{PA} relative to Alex? (c) What is the acceleration a_{PB} of car P relative to Bob during braking?

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Problem

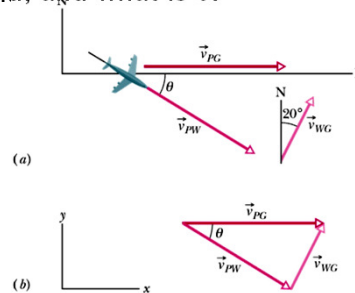
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$$\begin{aligned}
 \text{(a)} \quad & v_{PA} = v_{PB} + v_{BA} \\
 & -78 \text{ km/h} = v_{PB} + 52 \text{ km/h} \\
 & v_{PB} = -78 \text{ km/h} - 52 \text{ km/h} \\
 & \quad = -130 \text{ km/h} \\
 \text{(b)} \quad & v = v_0 + at \\
 & a = \frac{v - v_0}{t} = \frac{0 - (-78 \text{ km/h})}{10 \text{ s}} \left(\frac{1000 \text{ m/km}}{3600 \text{ s/h}} \right) \\
 & \quad = 2.2 \text{ m/s}^2 \\
 \text{(c)} \quad & v = v_0 + at \\
 & a_{PB} = \frac{v - v_0}{t} \\
 & \quad = \frac{-52 - (-130 \text{ km/h})}{10 \text{ s}} \left(\frac{1000 \text{ m/km}}{3600 \text{ s/h}} \right) \\
 & \quad = 2.2 \text{ m/s}^2
 \end{aligned}$$

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Problem

A plane moves due E while the pilot points the plane somewhat S of E, toward a steady wind that blows to the NE. The plane has velocity $\vec{v}_{PW}$ relative to the wind, with an airspeed, relative to the wind of 215 km/h directed at angle θ S of E. The wind has a velocity $\vec{v}_{WG}$ relative to the ground with a speed 65 km/h, directed 20.0° E of N. What is the magnitude of the velocity $\vec{v}_{PG}$ of the plane relative to the ground, and what is θ .

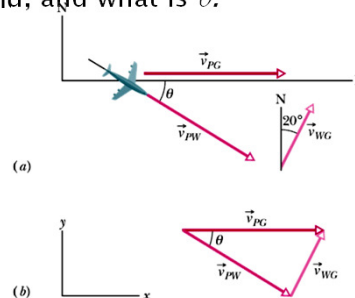


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Problem

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$$\begin{aligned}\vec{v}_{PG} &= \vec{v}_{PW} + \vec{v}_{WG} \\ v_{PG,y} &= v_{PW,y} + v_{WG,y} \\ 0 &= -215 \sin \theta + 65 \cos 20^\circ \\ \theta &= \sin^{-1} \left(\frac{65 \cos 20^\circ}{215} \right) = 16.5^\circ \\ v_{PG,x} &= v_{PW,x} + v_{WG,x} \\ &= 215 \cos 16.5^\circ + 65 \sin 20^\circ \\ &= 228 \text{ km/h}\end{aligned}$$



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